

CS 171 / EE 142: Intro to ML and DM

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Slide Set 8: Feature Construction



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
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 - ▶ For use only by enrolled students in the course

Non-linear Regression?

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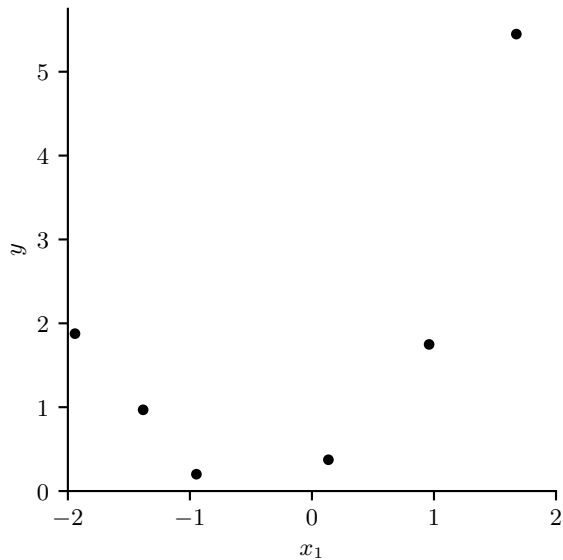
Non-linear Regression?

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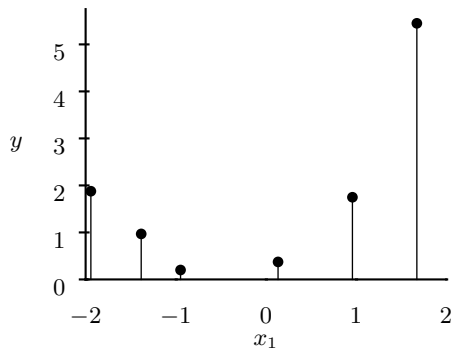
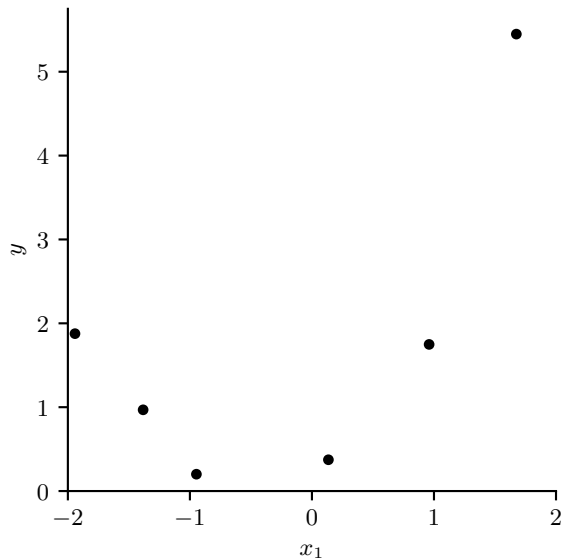
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So instead of just adding the “0th” attribute (always 1)
also add other attributes that can be calculated from the given attributes.

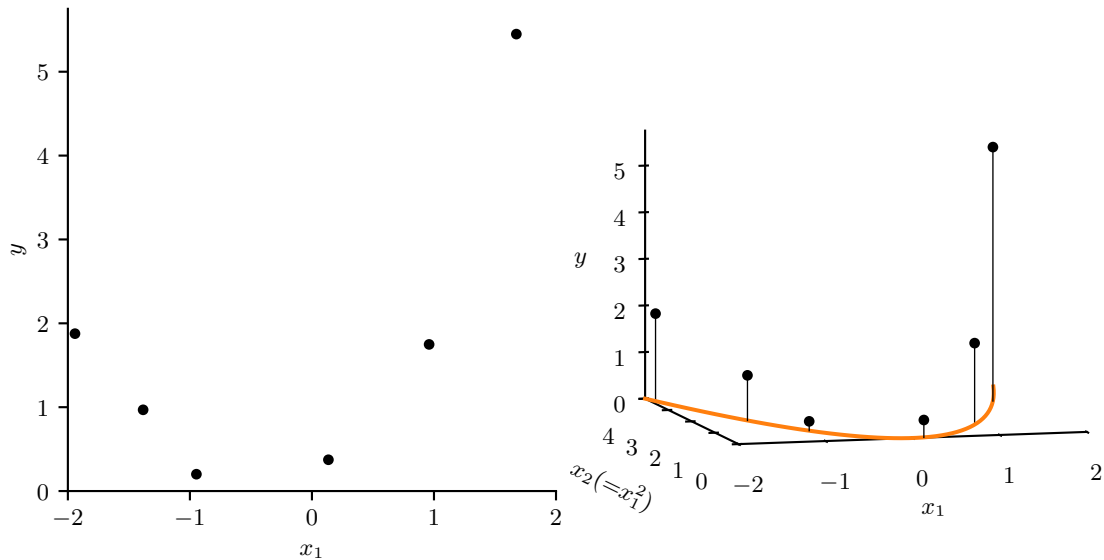
Non-linear regression?



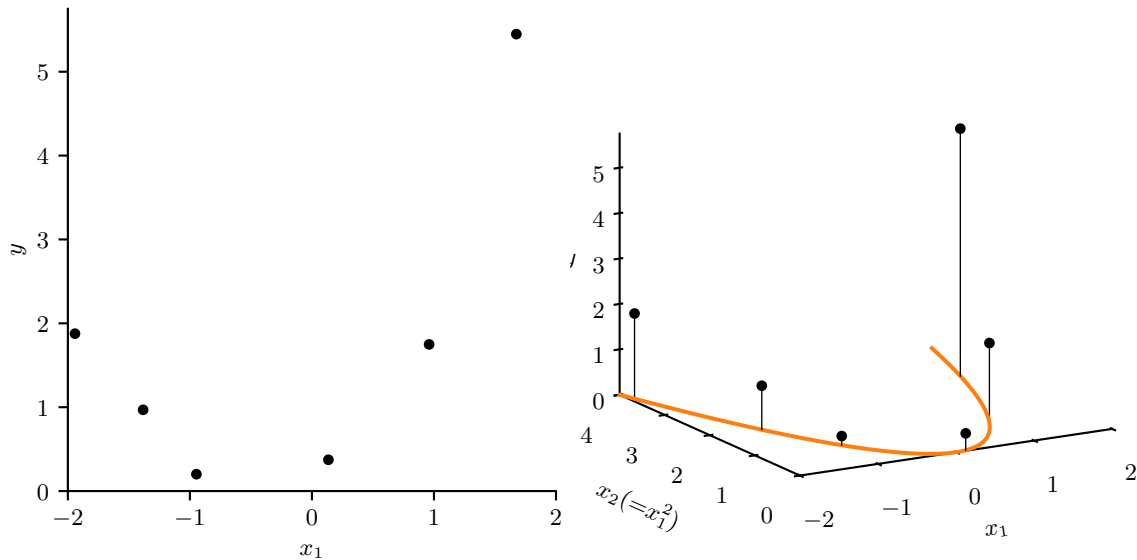
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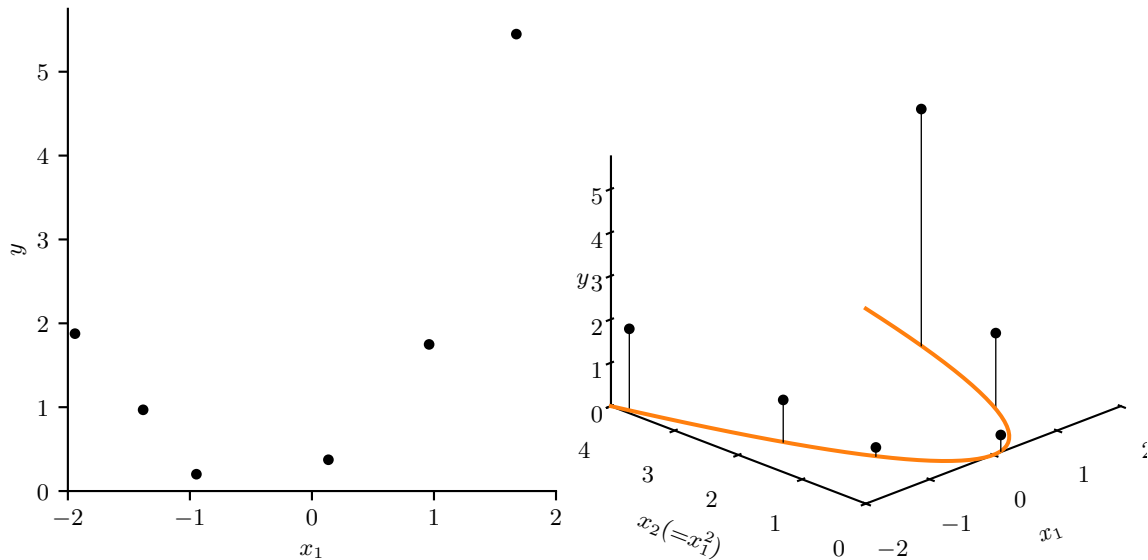
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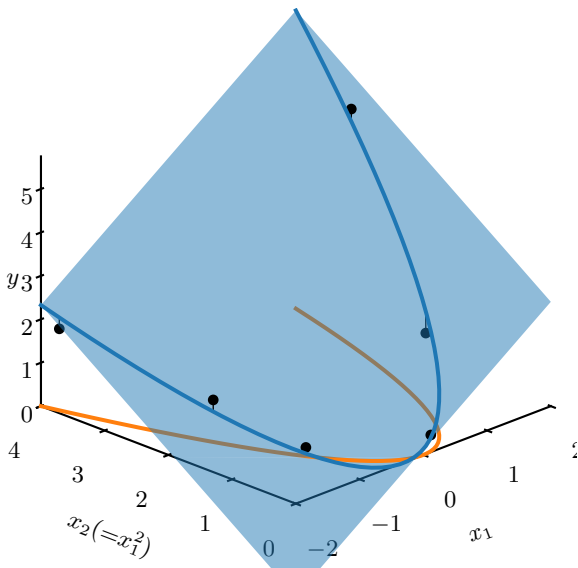
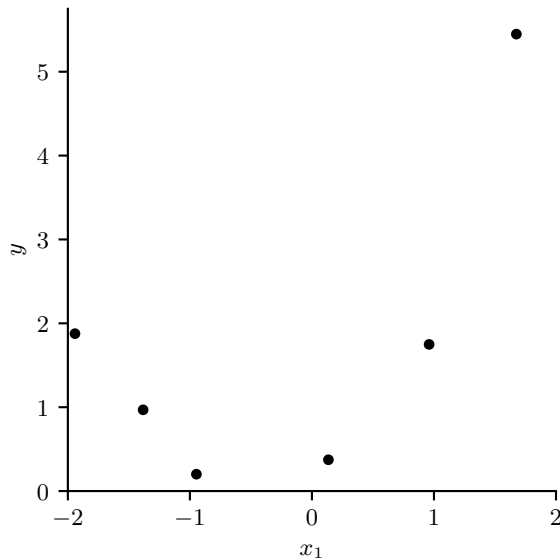
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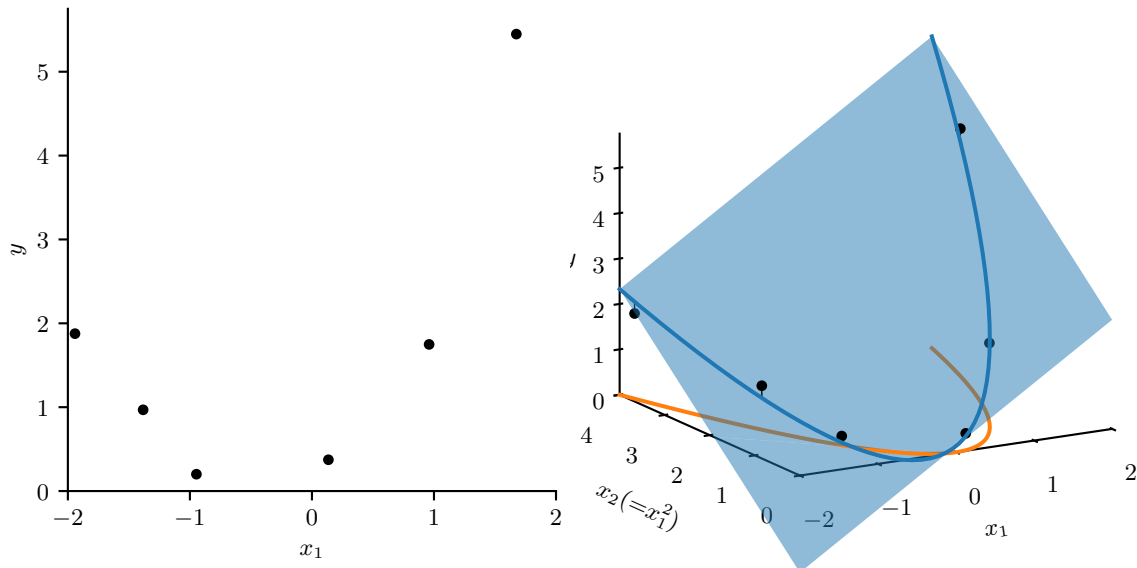
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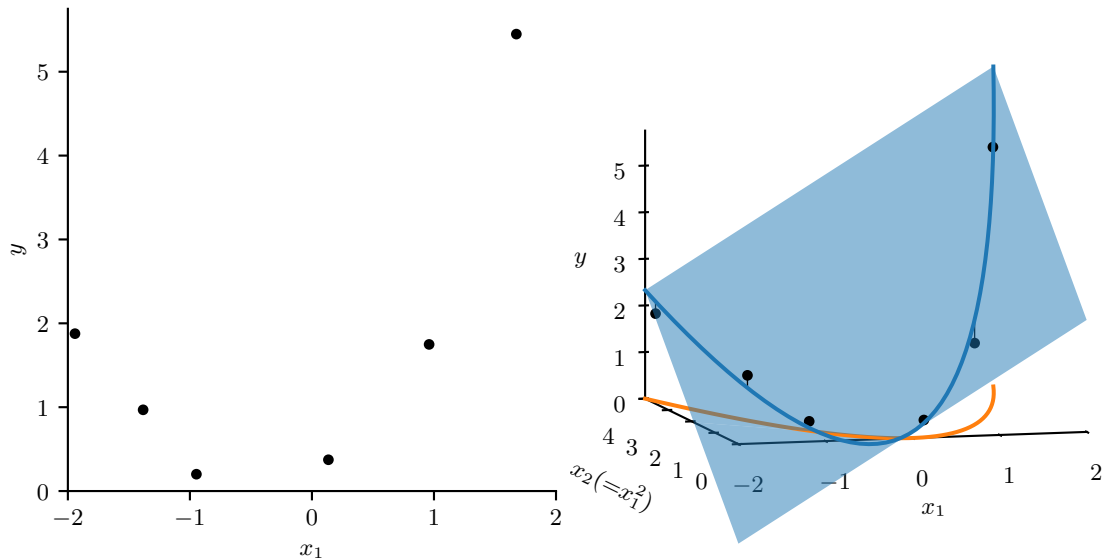
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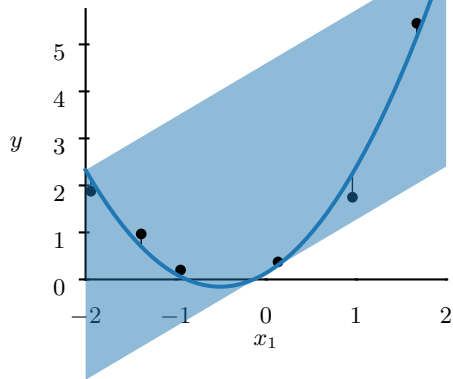
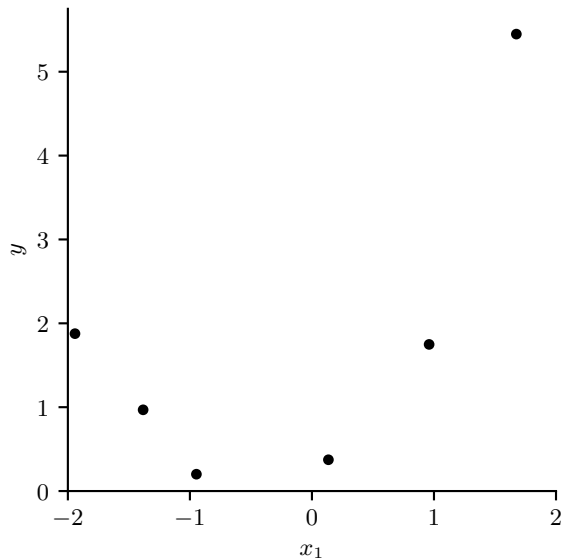
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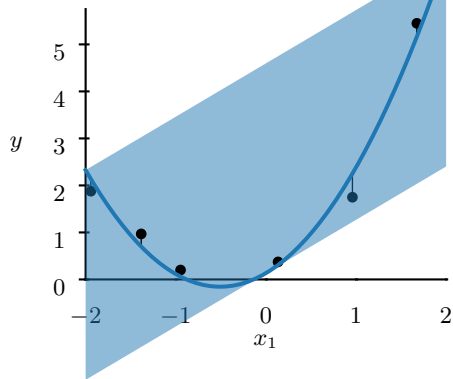
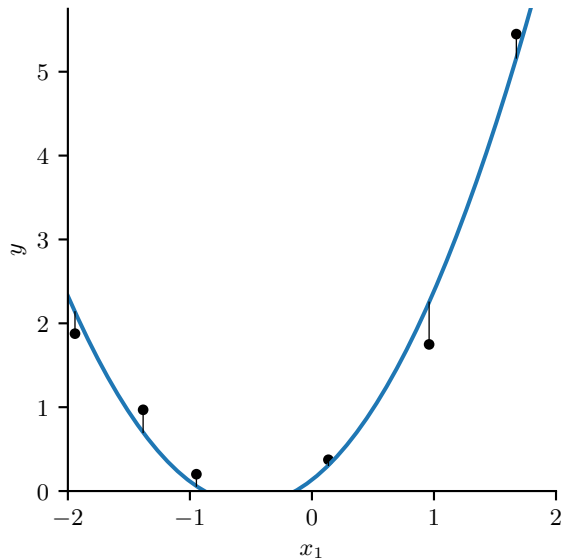
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Feature Mapping

The function that takes the raw attributes and creates features from them is often written $\phi(\vec{x})$.

For instance

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Instead of $\mathbf{X}$, we sometimes call the new matrix Φ :

$$\mathbf{X} = \begin{bmatrix} \text{---}\vec{x}_1^\top\text{---} \\ \text{---}\vec{x}_2^\top\text{---} \\ \vdots \\ \text{---}\vec{x}_m^\top\text{---} \end{bmatrix}$$

$$\Phi = \begin{bmatrix} \text{---}\phi(\vec{x}_1)^\top\text{---} \\ \text{---}\phi(\vec{x}_2)^\top\text{---} \\ \vdots \\ \text{---}\phi(\vec{x}_m)^\top\text{---} \end{bmatrix}$$

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The learning equation correspondingly changes notation:

$$\vec{w} = (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{Y}) \qquad \vec{w} = (\Phi^\top \Phi)^{-1} (\Phi^\top \mathbf{Y})$$

And of course the resulting function changes equation too:

$$f(\vec{x}) = \vec{x}^\top \vec{w} \qquad f(\vec{x}) = \phi(\vec{x})^\top \vec{w}$$

Feature Mapping Notes

- Original $\vec{x}$ can be called the raw features
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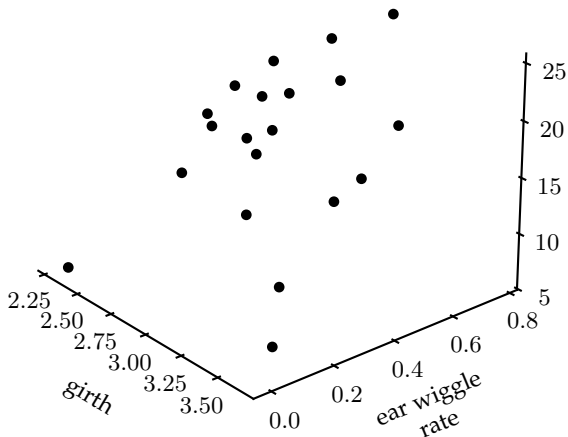
No.

yes	no
$w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_1x_2 + w_5x_2^2$	$w_0 + w_1 \log(w_2 + w_3x_1)$
$w_0 + w_1 \cos(2\pi x_1) + w_2 \sin(2\pi x_2)$	$w_0 + w_1 \sin(w_3x_1) + w_2 \sin(w_4x_2)$

Linear methods work if the functions in the hypothesis space are linear in $\vec{w}$.

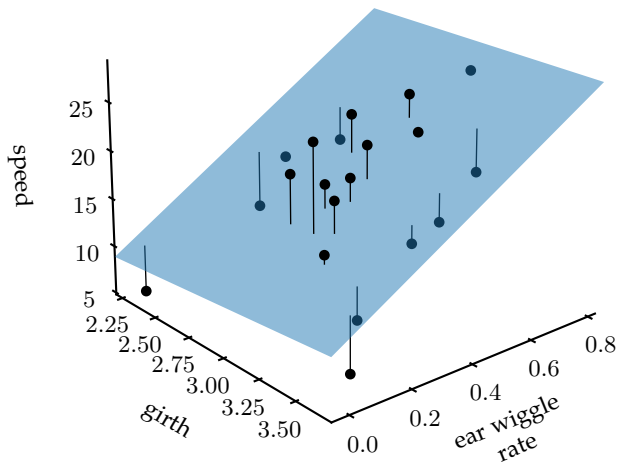
Overfitting with Features, Least Squares Regression

Data



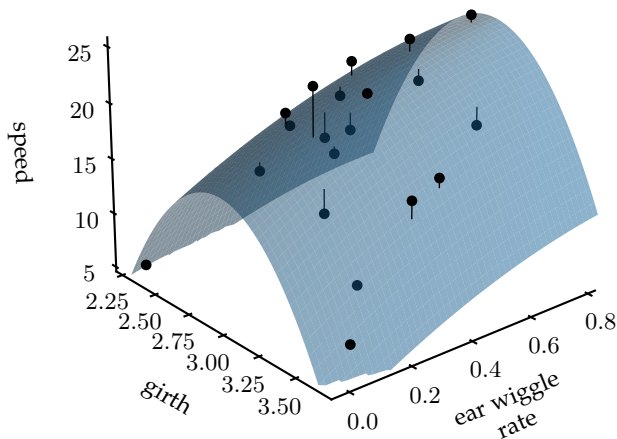
Overfitting with Features, Least Squares Regression

Linear: $\phi(x) = [1 \ x_1 \ x_2]^T$
 $n = 3$, mean sq. error = 15.55



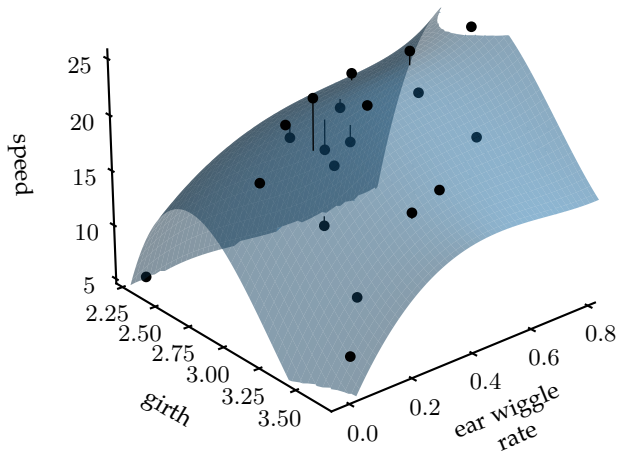
Overfitting with Features, Least Squares Regression

2nd order: $\phi(x) = [1 \ x_1 \ x_2 \ x_1^2 \ x_1x_2 \ x_2^2]^\top$
 $n = 6$, mean sq. error = 2.11



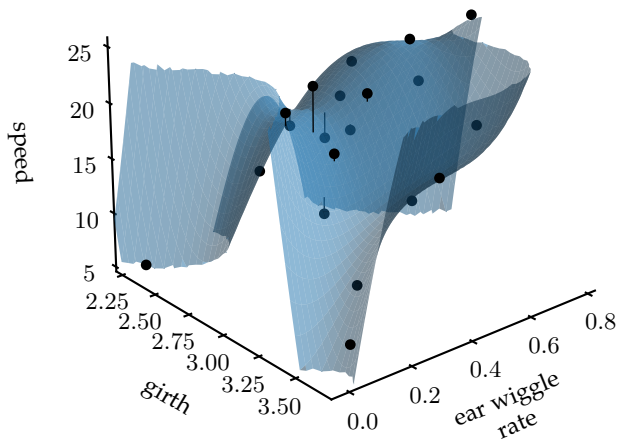
Overfitting with Features, Least Squares Regression

3rd order: $\phi(x) = [1 \ x_1 \ x_2 \ x_1^2 \ x_1x_2 \ x_2^2 \ x_1^3 \ x_1^2x_2 \ x_1x_2^2 \ x_2^3]^\top$
 $n = 10$, mean sq. error = 1.64



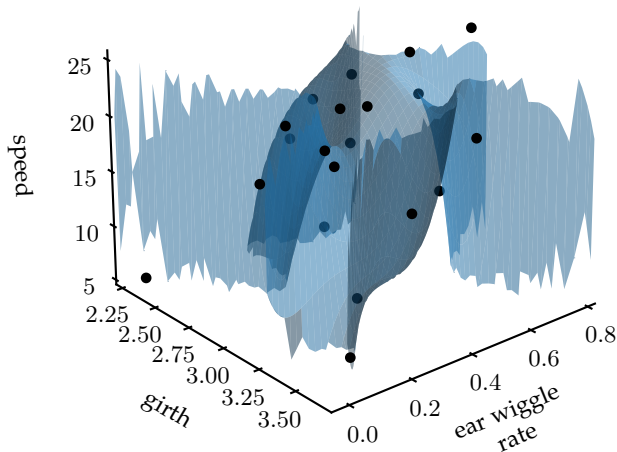
Overfitting with Features, Least Squares Regression

4th order: $\phi(x) = [1 \ x_1 \ x_2 \ x_1^2 \ x_1x_2 \ x_2^2 \ x_1^3 \ x_1^2x_2 \ x_1x_2^2 \ x_2^3 \ x_1^4 \ x_1^3x_2 \ x_1^2x_2^2 \ x_1x_2^3 \ x_2^4]^\top$
 $n = 15$, mean sq. error = 1.34

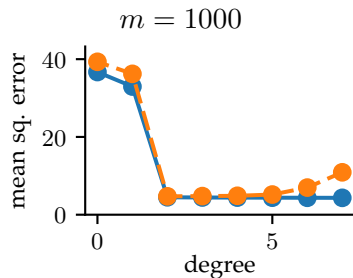
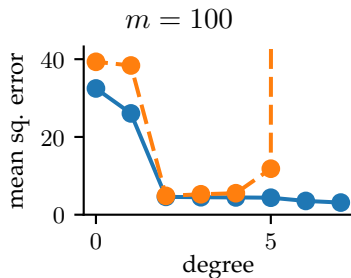
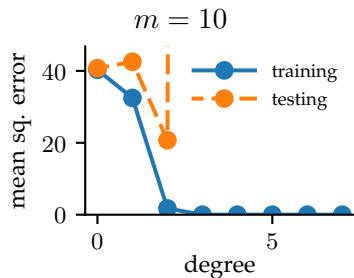


Overfitting with Features, Least Squares Regression

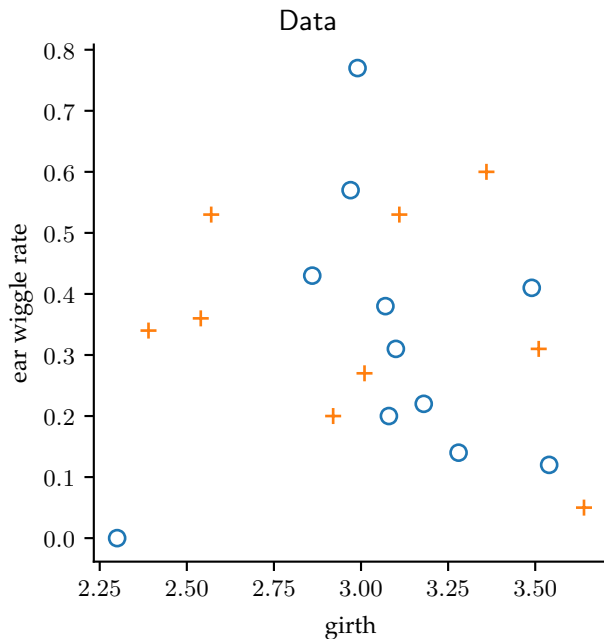
5th order: $\phi(x) = [1 \ x_1 \ x_2 \ x_1^2 \ x_1x_2 \ x_2^2 \ x_1^3 \ x_1^2x_2 \ \dots \ x_2^5]^\top$
 $n = 21$, mean sq. error = 2.25×10^{-6}



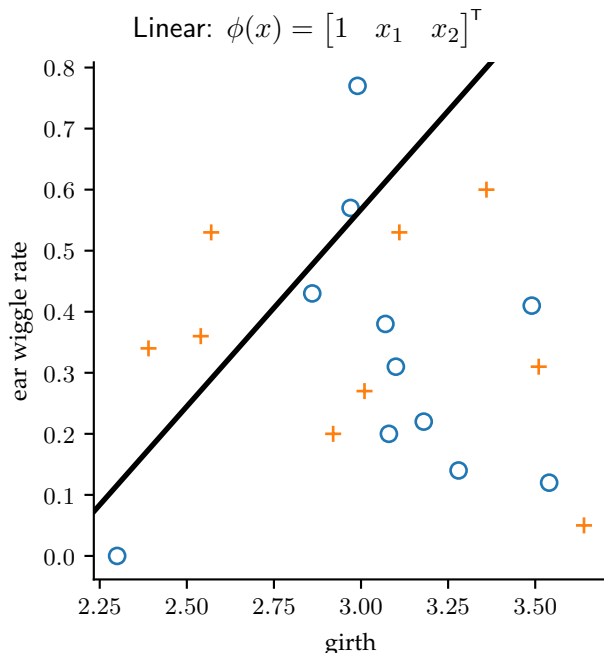
Overfitting with Features, Linear Regression



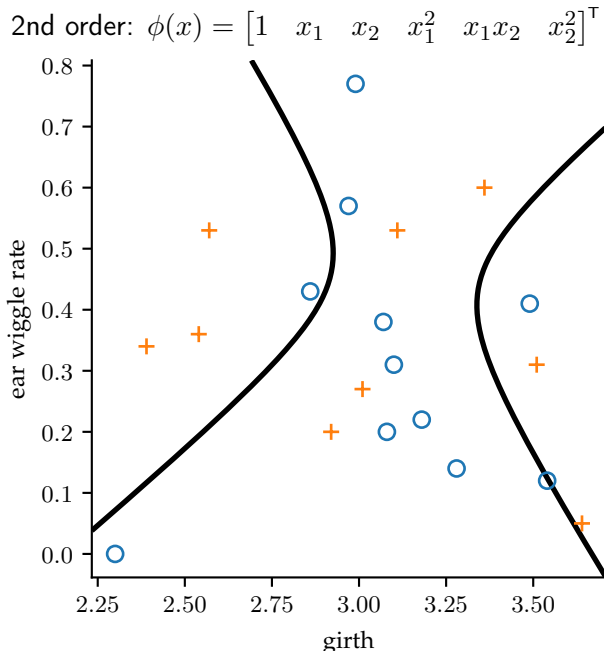
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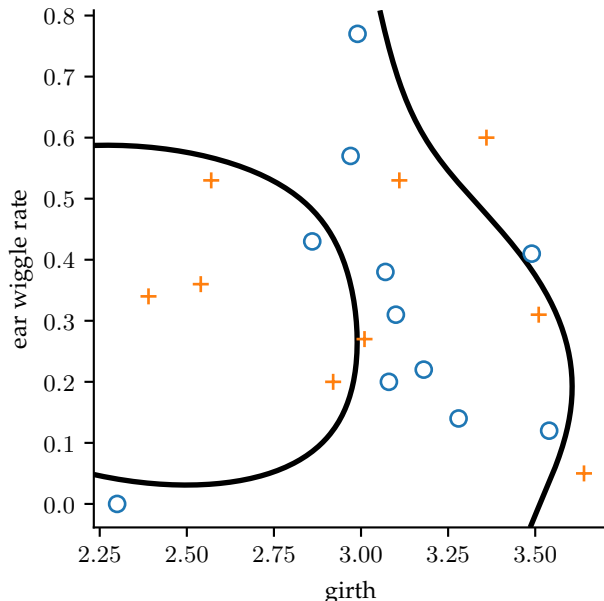


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